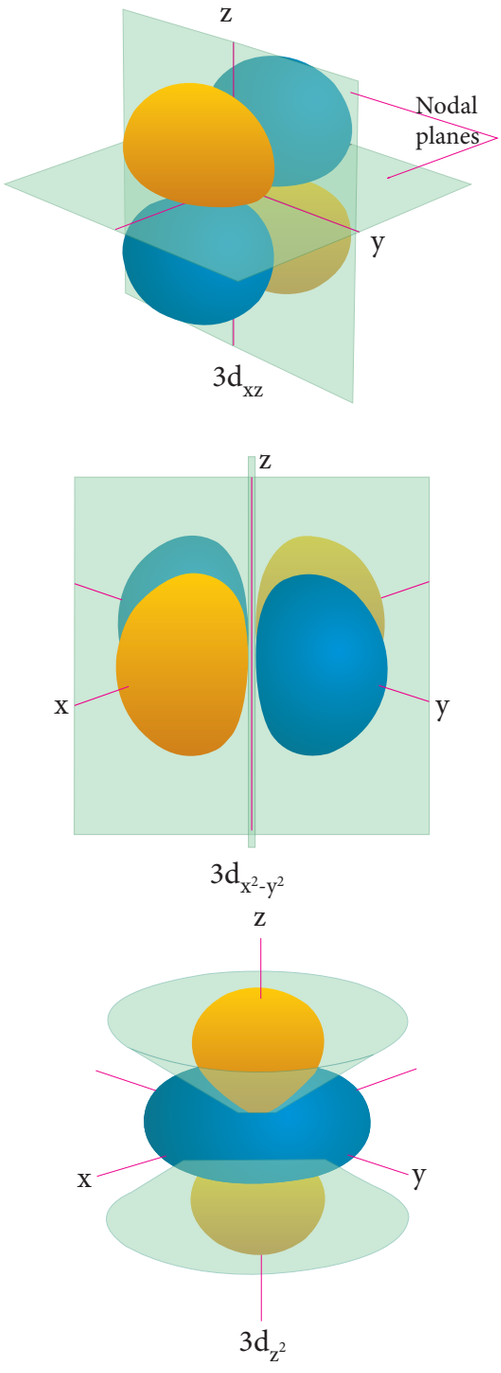


Figure 2.9 shapes of d orbitals

For 'f' orbital, $l = 3$ and the m values are -3, -2,-1, 0, +1, +2, +3 corresponding to seven f orbitals f_{z3}, f_{xz2}, f_{yz2}, f_{xyz}, $f_{z(x^2-y^2)}$, $f_{x(x^2-3y^2)}$, $f_{y(3x^2-y^2)}$, which are shown in Figure 2.10. There are 3 nodal planes in the f-orbitals.

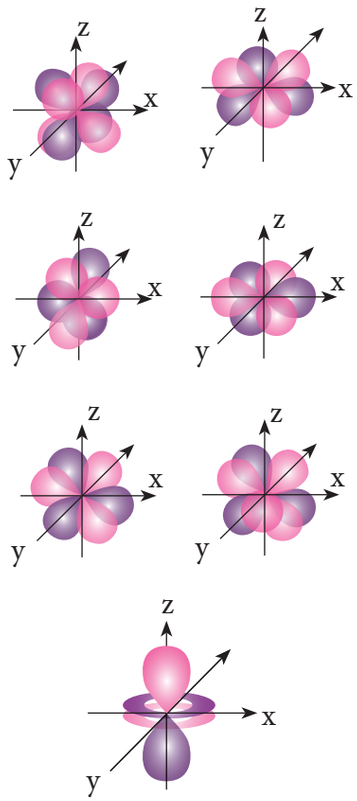


Figure 2.10 shapes of f-orbitals

Evaluate Yourself

4. Calculate the total number of angular nodes and radial nodes present in 3d and 4f orbitals.

2.5.2 Energies of orbitals

In hydrogen atom, only one electron is present. For such one electron system, the energy of the electron in the nth orbit is given by the expression

$$E_n = \frac{(-1312.8)\,Z^2}{n^2}\ \text{kJ mol}^{-1}$$

From this equation, we know that the energy depends only on the value of principal quantum number. As the n value increases the energy of the orbital also increases. The energies of various orbitals

11th Std Chemistry 038-067.indd 50 06-01-2021 15:06:51